

Energy management and sustainability assessment of renewable energy communities: The Italian context

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ABSTRACT

The Clean Energy for all Europeans Package introduced the now well-known Renewable Energy Community (REC), decentralized and democratic energy production and distribution configurations that are strategic to foster the integration of renewable sources in urban areas. The design phase of a REC is of utmost importance in terms of energy performance assessment and cost-effectiveness of the investment. In this study, an optimization model is presented to orient energy experts and urban planners in the capacity sizing and flow management of RECs under the Italian regulatory framework. Physical and virtual sharing schemes for centralized and decentralized configurations have been compared, also encompassing the chance to include storage systems. Results indicate that, the economic benefits for REC's participants increase with the amount of energy shared under the virtual scheme in a centralized distribution configuration. Moreover, purpose-built sustainability indicators highlighted the significant reduction of carbon emissions, up to 34 % compared to the traditional scenario, and a reduction of energy poverty, with 14 low-income families included in the REC.

1. Introduction

According to the definition of the Directive 2018/2001, a Renewable Energy Community (REC) is a legal entity “based on open and voluntary participation with the primary purpose of providing economic, environmental or social community benefits for its members, stakeholders and for the areas where it operates” [1]. The members of RECs produce renewable electricity for their consumption, and may store or sell the self-produced renewable electricity either to the free electricity market or to other members of the same energy community; in light of all these listed characteristics, they are usually referred to as “self-consumers” or “prosumers” [1]. In Italy, energy communities are subject to the technical rules outlined by the GSE, the Italian Energy Services Operator [2], and by the regulation 318/2020/R/EEL of the ARERA, the Italian Regulation Agency for Environment, Network and Energy [3]. In particular, the former gives insights into the modes for which the electricity produced within the REC should be self-consumed, shared among members, or exchanged with the national power grid while the latter defines the incentives and economic benefits associated with each electricity flow. As a further element, the document introduces a new virtual scheme for energy sharing among members. Under this scheme, the electricity produced by the energy community can be virtually

consumed to cover also the electrical demands of members not physically connected to the production system. One of the main advantages of choosing a virtual scheme consists in the avoiding of a dedicated distribution network construction, with added technical constraints and high investment costs [4]. This is one of the reasons why virtually shared energy is incentivized from the normative.

1.1. Literature review

The topic of energy communities is attracting attention from researchers and local stakeholders, such as mayors, energy planners, and citizens. In the following, an overview of the most recent contributions is presented, with a focus on Italian regulation and specific Italian case studies, technologies and methods. When tackling the design phase of an energy community, one of the most critical aspects is related to the optimal configuration of technologies and energy-sharing strategies among members. Given the fact that different incentives are associated with different energy flows, the future performance of the community is highly affected by the production systems to be installed. During the design phase, information on the size of the energy technologies, for both production and storage, and their optimal location within the geographical boundaries of the community is strategic to plan cost-effective investments. The scientific literature is rich in examples and

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Nomenclature*Acronyms*

REC	Renewable Energy Community
ESS	Energy Storage System
PV	Photovoltaic panel
PSC	Physical Self-Consumption
VSC	Virtual Self-Consumption
UoM	Unit of Measure

Sets

$b \in \{1, 2, \dots, B\}$	Member of the REC
$t \in \{1, 2, \dots, 8760\}$	Time-step
$y \in \{1, 2, \dots, Y\}$	Year in which the economical flows are evaluated

Parameters

B	Set of members of the REC
Y	Set of years of the investment horizon
DEM_{tb}^{en}	Electrical demand of member b at time-step t [kWh]
en_t^{prod}	Electrical energy produced by PVs at time-step t , normalized by the installed nominal power [kWh/kW _n]
X_b	Binary vector: 1 if member b is physically connected to the production technology, 0 otherwise
PWR_b^{max}	Maximum nominal power that can be installed on the rooftop of member b [kW _n]
$p^{ESS,max}$	Maximum charged/discharged power of the ESS within one time-step [kWh]
η^{in}	Charging efficiency of the ESS [%]
η^{out}	Discharging efficiency of the ESS [%]
Dod	Depth of discharge of the ESS as a percentage of the total capacity [%]
cap_b^{ESS}	Total capacity of the ESS installed for member b [kWh]
$capex^{prod}$	Capital expenditures of production technology, normalized by the nominal power [€/kW _n]
$opex^{prod}$	Annual operational expenditures for the production technology, normalized by the nominal power [(€/y)/kW _n]
$capex^{ESS}$	Capital expenditures of ESS, normalized by the total capacity [€/kWh]
$opex^{ESS}$	Annual operational expenditures of ESS, normalized by the total capacity [(€/y)/kWh]
Inc	Incentive tariff for the electrical energy that is virtually shared within the REC [€/kWh]
Ncr	Network charges restoration for the electrical energy that is virtually shared within the REC [€/kWh]
buy^{grid}	Unitary cost for buying electrical energy from the national power grid [€/kWh]
$sell^{grid}$	Unitary price for selling electrical energy to the national

	power grid [€/kWh]
Disc	Annual discount rate [%]

Continuous variables

PWR_b	Installed nominal power of production technology physically connected to member b [kW _n]
EN_{tb}^{prod}	Electricity produced by production technology physically connected to member b at time-step t [kWh]
EN_{tb}^{PSC}	Electricity instantaneously used for physical self-consumption by member b at time-step t (only for scenarios without ESSs) [kWh]
$EN_{tb}^{PSC,ESS}$	Electricity that is used for physical self-consumption by member b at time-step t (only for scenarios with ESSs) [kWh]
$EN_{tb}^{surplus}$	Electricity surplus of member b at time-step t [kWh]
$EN_{tb}^{deficit}$	Electricity deficit of member b at time-step t [kWh]
$EN_t^{released}$	Electricity released by the REC to the national power grid at time-step t [kWh]
EN_t^{drawn}	Electricity drawn from the national power grid by the REC at time-step t [kWh]
EN_t^{VSC}	Electricity instantaneously used for virtual sharing within the REC at time-step t (only for scenarios without ESSs) [kWh]
$EN_t^{VSC,ESS}$	Electricity used for virtual sharing within the REC at time-step t (only for scenarios with ESSs) [kWh]
EN_{tb}^{charge}	Electricity used to charge the ESS physically connected to member b at time-step t [kWh]
$EN_{tb}^{discharge}$	Electricity discharged from the ESS physically connected to member b at time-step t [kWh]
SoC_{tb}	State of charge of the ESS physically connected to member b at time-step t [kWh]
EN_t^{imp}	Electricity imported from the national power grid to the REC net of the electricity virtually shared among members at time-step t [kWh]
EN_t^{exp}	Electricity exported from the REC to the national power grid net of the electricity virtually shared among members at time-step t [kWh]
NPV	Net Present Value [€]
CF_y	Cash flow for year y [€/y]
INV	Investment costs [€]
SAV_y	Savings obtained from physical self-consumption during year y [€/y]
REV_y	Revenues obtained from energy sharing and selling to the national power grid during year y [€/y]
$COST_y$	Operational costs during year y [€/y]

contributions in this direction. A significant number of papers focuses on the design and operation phase of energy communities, with some specific differences related to the set of considered technologies. In this case, optimization modelling approaches are the most largely diffused. Usually, the optimization is carried out employing mathematical programming, e.g., linear programming (LP) or mixed integer linear programming (MILP) models. More in detail, in the work of [5], a multi-criteria optimization model is elaborated to size a PV system equipped with storage for an energy community in Italy. The goal of the model was to minimize the electricity exchanged with the power grid and maximize the amount of shared electricity. A case study of an Italian energy community is also presented in Coletta and Pellegrino [4]. Here, the authors presented an optimization model for the maximization of the NPV and considered centralized and distributed PV configurations with

centralized batteries. In Viti et al. [6], a techno-economic comparison between two different scenarios where citizens act as single self-consumer or as an energy community is performed, again under the Italian regulation. From the technological viewpoint, PV, heat pumps and combined heat and power (CHP) units are usually the most diffusely adopted renewable technologies in the context of urban environments, given their potential of being easily integrated into buildings (or in proximity to buildings) even with few available spaces [7–9]. As an application, the research of [10] described four possible technical arrangements for PV on apartment buildings and demonstrated that self-sufficiency and self-consumption increased thanks to the aggregation of different loads. Configurations with PV are also tackled in the work of [11]. The authors presented a framework to optimize PV-based energy communities, using a Monte Carlo approach to simulate the electrical

demands of the community. Although hybrid renewable energy systems can frequently outperform conventional configurations, managing the design phase of such a complex system is more challenging. An overview of the mostly diffused optimization methods and novel trends for the optimal design is discussed in Nehrir et al. [12]. Focusing on storage systems, the concept of load shifting and demand side management for energy communities have been applied with a multi-year sizing methodology [13]. Energy management in communities is tackled in Tostado-Véliz et al. [14], with a model considering controllable demand, vehicle-to-grid, and energy storage. The relevance in choosing the optimal storage system size and type is stressed by Guo et al. [15], who proposed a two-layer co-optimization method tailored to a distributed energy system combining lithium batteries for power-to-heat and power-to-cold. The paper of [16] presented an optimization model able to account for various energy storage capacities and different loads, namely residential and commercial. Emerging technologies based on hydrogen are also being employed successfully in the energy sector, particularly in remote places where electrification is not possible due to a lack of local power infrastructure. A thorough techno-economic analysis of the PV-hydrogen energy system to electrify rural houses in Pakistan was carried out using HOMER software [17]. One of the most relevant findings demonstrates that, if the grids are located beyond a specific limit, the suggested energy system is more cost-effective than grid extension. Hydrogen energy storages are also proving to be an efficient alternative to reduce seasonal mismatches between energy supply and cooling, heating, and electrical demands of energy communities. An example in this direction is offered by the work of [18], where a multi-variable multi-objective mixed integer nonlinear programming (MILNP) is adopted to select the optimal capacity of hydrogen energy storage systems based on the evaluation of energy, economic, environmental, and grid dependence. A MILP model is also formulated in Braeuer et al. [19] for integrating multi-energy systems in an energy community with multi-family buildings among its members.

The technological design perspective needs to be aligned with energy performance issues and the cost-effectiveness of the investments. A significant part of the literature, indeed, tackles economic aspects beyond the pure dimensioning of renewable systems. As a point of discussion, one of the main economic aspects that has to be considered during the management phase of energy communities regards the cash flows for all members. To address this topic, Zatti et al. [20] proposed a Shapley value-based approach to distribute a part of the community's incomes to members and results demonstrated the effectiveness of this management scheme, also bringing the additional advantage of encouraging members towards energy-efficient behaviors. A prediction model grounded on the evolutionary game theory is used in Cheng et al. [21] to look at the equilibrium characteristics of the long-term strategic bidding practice for the generation-side markets. The same approach is taken in Cheng et al. [22] to address the demand-side management issue. Operational strategies need to be carefully evaluated in terms of their ability to reduce costs, especially when dealing with high RES penetration. This was accomplished in Zhang et al. [23] by applying a unique power scheduling method tailored for microgrids with distributed generation and storage. Another aspect that needs to be carefully investigated, before the constitution of an energy community, is the way citizens should be selected to form the energy community. The work of [24] deepened this aspect, and the potential prosumers' influence on the final performance of the community is evaluated using the MILP technique. The authors proved that is possible to reduce costs in the range of 15–20 % when tackling the aggregation problem using optimization. Similarly, Sudhoff et al. [25] focussed their analysis on the correlation between grid-friendly operation of flexible assets, economic benefits and sharing among members in rural, suburban and urban grid topologies. Some researchers have recognized the importance of improving the integration of energy communities with power grid systems by proposing novel data-driven flexibility models to optimize the energy production and consumption of the energy community members [26]. A

similar viewpoint is presented in Waseem et al. [27], suggesting a novel framework for scheduling home appliances in the presence of real-time price signals to reduce electricity costs and the peak-to-average ratio. The energy community forming process can also be tackled as done by Barbour et al. [28], who developed a methodology to create local smart energy communities considering the monthly consumption of potential members and their geographical locations. The ability of artificial intelligence techniques, such as machine learning, to make accurate predictions has recently opened up new possibilities for addressing a variety of energy-related issues. In this direction, Cheng and Yu [29] conducted a review of the applications of seven representative machine-learning techniques. The possible use of each technique in electric power and smart energy systems is here thoroughly discussed. Sustainability indicators specifically tailored to energy communities have been reviewed in Gjorgievski et al. [30]. However, as also stressed by the authors, a quantitative framework for evaluating the social impacts and benefits is lacking in the literature. Regarding social aspects, Lode et al. [31] aimed to investigate the correlation of specific socio-demographic factors with the existence of energy communities. Finally, a social and environmental assessment is performed for an energy community with solidarity targets in Kaiser et al. [32]. From the conducted literature review, it emerges that the RECs can be studied and analyzed from various perspectives and, in addition, national legislations bring different approaches and management schemes. Often, urban planners, energy manager and, generally, any stakeholder involved in the constitution process of a REC encounter barriers related to the optimal capacity to be installed, the optimal number of members and the energy consumption profiles to be chosen (as well as the final use of buildings – residential, commercial or industrial), the optimal operation and management of the energy flows across the community able to guarantee higher revenues and cost savings. Therefore, translating normative indications into modelling procedures, tailored to the national regulations and targets can pave the way for a more rational and sustainable development of RECs into the national boundaries. For this reason, the focus of this paper is on Italian RECs. This research proposes an optimization model to provide insights on the energetic, economic and social impact of RECs at varying electrical distribution schemes, i.e., distributed or centralized, ranges of PV penetration, installation of distributed energy storage systems. In addition to the previously mentioned questions, other aspects need to be tackled, such as the performance assessment and monitoring of RECs after their constitution. To this scope, purpose-built indicators could offer an holistic overview on the economic, environmental and social impact of the RECs to help members and stakeholders during any decision-making process. Under these premises, this paper aims at supporting REC members, local authorities, energy planners and any stakeholders in all the stages of an energy community constitution: from the definition to the design and implementation.

The remainder of this paper is structured as follows. Section 2 introduces the concept of renewable energies community given by main European Directives and how this concept has been transposed in the Italian context. The optimization model, its mathematical formulation, assumptions, and different configuration scenarios are then presented in Section 3. In Section 4, the analyzed case study is presented and the discussion of results deriving from the optimized scenarios is reported in Section 5. Finally, Section 6 provides the concluding remarks of the study.

2. Overview of the Italian normative for energy communities

As introduced in the previous paragraph, energy communities are open and voluntary legal entities with economic, environmental, and social objectives, and are constituted by consumers, local authorities, and small-medium enterprises. This general definition is held in common with three different European Directives. Indeed, depending on key characteristics or activities of the energy community, it is possible to distinguish between *Renewable Energy Community*, as from Directive

2018/2001 [1], and *Citizen Energy Community* (CEC), from Directive 2019/944 [33], including also that form of CEC that falls into the Gas Market Directive 2009/73 [34]. Although having several features in common, REC and CEC need to be modelled and studied according to their specificities. For example, REC and CEC differentiate in terms of geographical scope (RECs are subjected to a proximity criterion), participation of large-scale stakeholders with economic interests (the membership of REC is restricted to actors whose participation is not related to their primary economic activity), and control (small enterprises located in the proximity of the geographical boundary of the REC can have control duties, not the same for CEC).

This study strictly refers to REC. Among the different key aspects that can be taken into consideration, the ownership and control of the produced energy is probably one of the most impacting novelties introduced by Directive 2018/2001. Indeed, REC can use and share the energy produced from its owning generation assets inside the community. At this point, it is worth explaining how the concepts of “*energy self-consumption*” and “*energy shared*” are conceived from the Italian regulation. If REC members own renewable production technologies, such as PV panels installed on their roofs, they gain the chance to locally self-consume renewable energy thanks to a physical connection to generators obtained through electrical cables. This scheme of self-consumption is referred to as “*physical self-consumption*”, PSC. Only after the fulfilment of their electrical demands, surplus energy from production systems’ owners can be virtually shared with other members of the community and the shared energy is often referred to as “*virtual self-consumption*”, VSC. It is, however, worth underlining that the energy used for virtual sharing is exchanged only between each member of the REC and the national power grid, but accounted for as energy from the REC. In this sense, the grid can be considered a sort of reservoir, without the costs of transmission and distribution. Therefore, the sum of VSC and PSC represents the total amount of energy self-consumed by the REC. To quantify the amounts of energy shared in both PSC and VSC schemes on an hourly basis, two different meters are needed: a private one for PSC (reading physical self-consumption, i.e., the surplus energy available for VSC) and a public one for VSC computation (reading the amount of energy shared among the REC members associated with the exceeding renewable production after PSC and using the grid as a tank for balancing the energy flows) [2,35]. As emerge for the case of VSC, the Italian normative considers the concepts of “*self-consumption*” and “*energy sharing*” as synonyms, since energy supplied from the grid is computed as part of the energy produced by the REC (and, therefore, self-consumed by the REC or shared among the REC members), but differentiated from the energy physically consumed by the owner of the production system.

REC may also be distinguished according to the sharing configurations, among which the most diffused are the distributed and centralized one. In the distributed configuration, each production technology is *physically* connected to one or more REC members (for each member, PSC applies) and, generally, the other REC members may benefit from the self-consumption according to the VSC scheme. More specifically, this is the case for which each REC member equipped with the production system satisfies its energy demand as PSC, and, eventually, if an energy surplus occurs, it can be used to cover the other REC members’ demands, as in the VSC scheme. For the centralized configuration, only VSC is allowed, and the self-consumption scheme is deemed equivalent to the VSC for all REC members.

The Italian normative gives also indications on how VSC should be computed for the RECs. This calculation results in the Eq. (1) below:

$$EN_h^{VSC} = \min_h \{ EN_h^{\text{released}}, EN_h^{\text{drawn}} \} \quad (1)$$

According to this, the amount of energy EN_h^{VSC} that is virtually shared within the REC is calculated as the minimum between the total energy released to the national power grid, EN_h^{released} , and the total energy drawn from it, EN_h^{drawn} , on an hourly basis. It is worth noting that, in the

centralized configuration, the entire production from renewable systems is released to the national power grid, since no PSC occurs. In this case, the amount of energy referred to as self-consumed from the REC is recognized only under the VSC scheme. Fig. 1 gives a schematic overview of the energy flows occurring in these two configurations.

Finally, the Italian normative also allows the introduction of Energy Storage Systems (ESSs) to increase the rates of PSC and VSC within the REC. ESSs enable the disaggregation between supply and load profiles, hence increasing load matching and economic benefits.

From the economic viewpoint and to quickly scale up the constitution of RECs, two incentives for energy shared under the VSC scheme are defined. The first consists of a valorization of 110 €/MWh for the renewable energy virtually shared within the community [36] and the second is a network charge restoration of 8.7 €/MWh to account for distribution losses avoided thanks of self-consumption [2]. It is worth noting that economic benefits from incentives are distributed among members according to private contracts agreed upon before the constitution of the REC [2]. However, in this paper cash flows are assumed as equally distributed among the REC members and all the renewable energy produced within the REC and released to the power grid can be sold in the free electricity market [2].

3. Materials and methods

The proposed optimization model aims at the design and the management of the energy flows across of a candidate REC, as described in the flowchart of Fig. 2. The initial step consists in the identification of the number of participants of the REC, both in terms of typology of buildings and number of consumers, as well as the identification of the physical boundaries of the community, necessary to verify administrative permissions and proximity to the medium voltage electrical cabin. The model is characterized by the definition of parameters and decision variables. Parameters inform the model on technical, environmental and economic data, such as the hourly consumption and production profiles of consumers, the roofs’ top areas, emission factors of the chosen technology compared to a reference traditional case, capital and operating costs of the investment. More in detail, the upper limit on the technology capacity is determined by the total space availability on the rooftop of the building in which the PV is installed. Emission factors, capital and operating costs are gathered for each technology to compute the environmental and economic performances of the REC. Decision variables are categorized into design and operation variables. While the latter refers to the energy exchanges among REC members, technologies, and the power grid, the former relates to the capacity of the production and storage systems. The equations of the model include operational constraints, capacity constraints, and an objective function, here formulated to maximize the net present value (NPV). The optimization model runs in MATLAB R2021a [37] for different energy-sharing schemes (centralized vs decentralized and with/without the insertion of ESSs) and different PV penetration strategies. Finally, a post-optimal analysis is performed to compare the economic, environmental, and social performances of the identified scenarios with the aim of tailored KPIs.

The optimization model presented in this section takes inspiration from [4,20], but also adds constraints and balances related to PV and ESSs and differentiates between distributed and centralized configurations. The objective function aims at maximizing the Net Present Value (NPV) that includes the contributions from the economic valorization, i.e., incentives and savings in the energy bills.

The mathematical model is written using MATLAB programming language [37]. Equations refer to both the distributed and centralized configurations and have been written following an hourly time step. For the scenarios including ESSs, the maximum capacities have been calculated in relation to the PV sizes. Equations are reported for both the distributed and centralized configuration.

The total energy production of each member b at time t is calculated as in Eq. (2) and Eq. (3), for the distributed and centralized

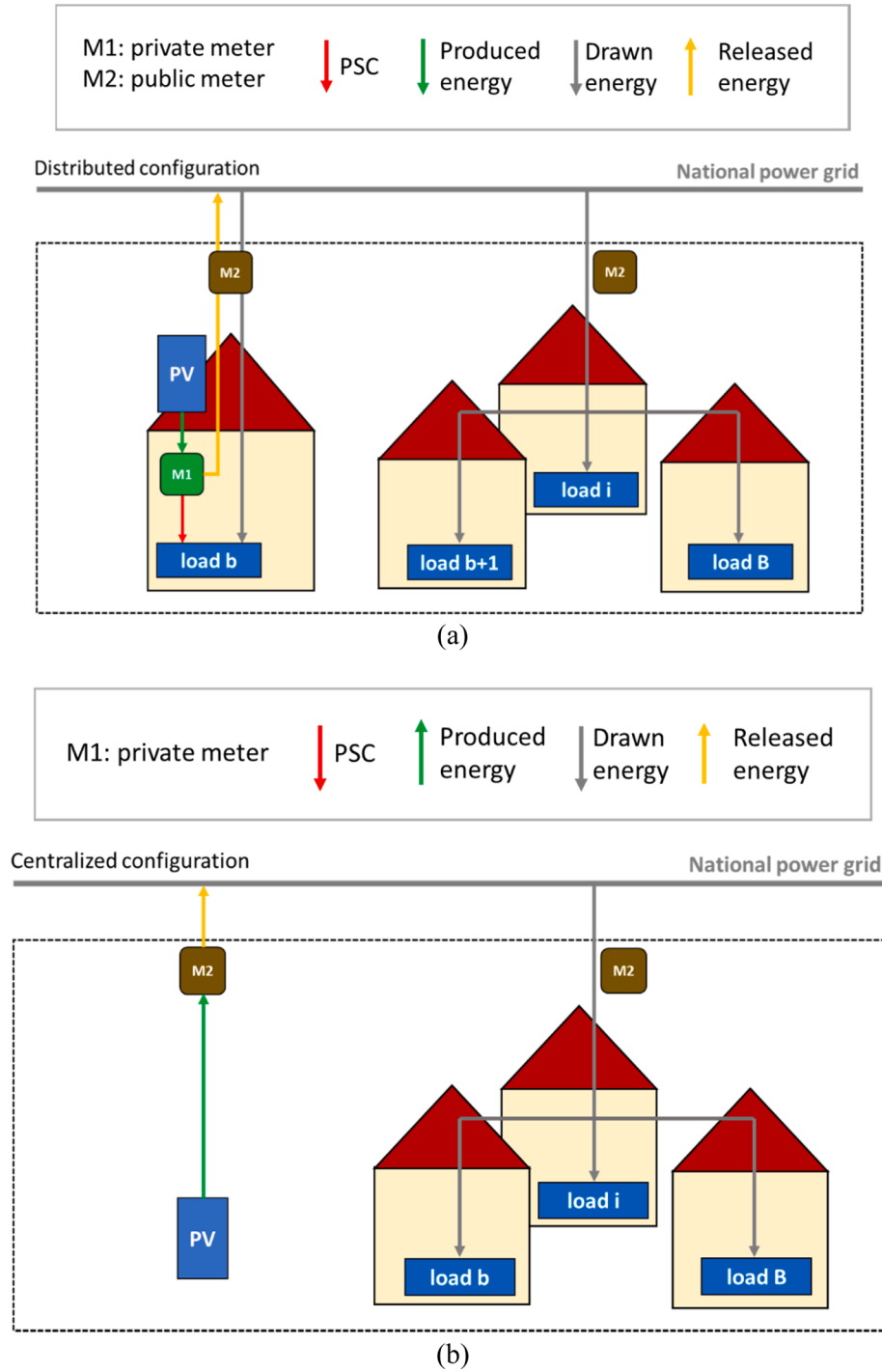


Fig. 1. (a) Distributed and (b) centralized energy sharing configurations.

configuration, respectively. For the distributed configuration in Eq. (2), the total energy production EN_{ib}^{prod} is calculated considering the installed PV nominal power and the number of members b physically connected to the PV panels, expressed using the binary vector X_b . For the centralized configuration, formulated in Eq. (3) the total energy production is calculated without reference to the physical connection between the production technology and the REC members.

Distributed configuration

$$EN_{ib}^{\text{prod}} = \frac{en_i^{\text{prod}} \cdot PWR_b \cdot X_b}{\sum_b X_b} \quad \forall t, b \quad (2)$$

Centralized configuration

$$EN_i^{\text{prod}} = en_i^{\text{prod}} \cdot PWR \quad \forall t \quad (3)$$

Eq. (4) and Eq. (5) express upper-bound constraints on the PV size based on the available rooftop area of each REC member.

Distributed configuration

$$PWR_b \leq PWR_b^{\text{max}} \quad \forall b \quad (4)$$

Centralized configuration

$$PWR \leq \sum_b PWR_b^{\text{max}} \quad \forall b \quad (5)$$

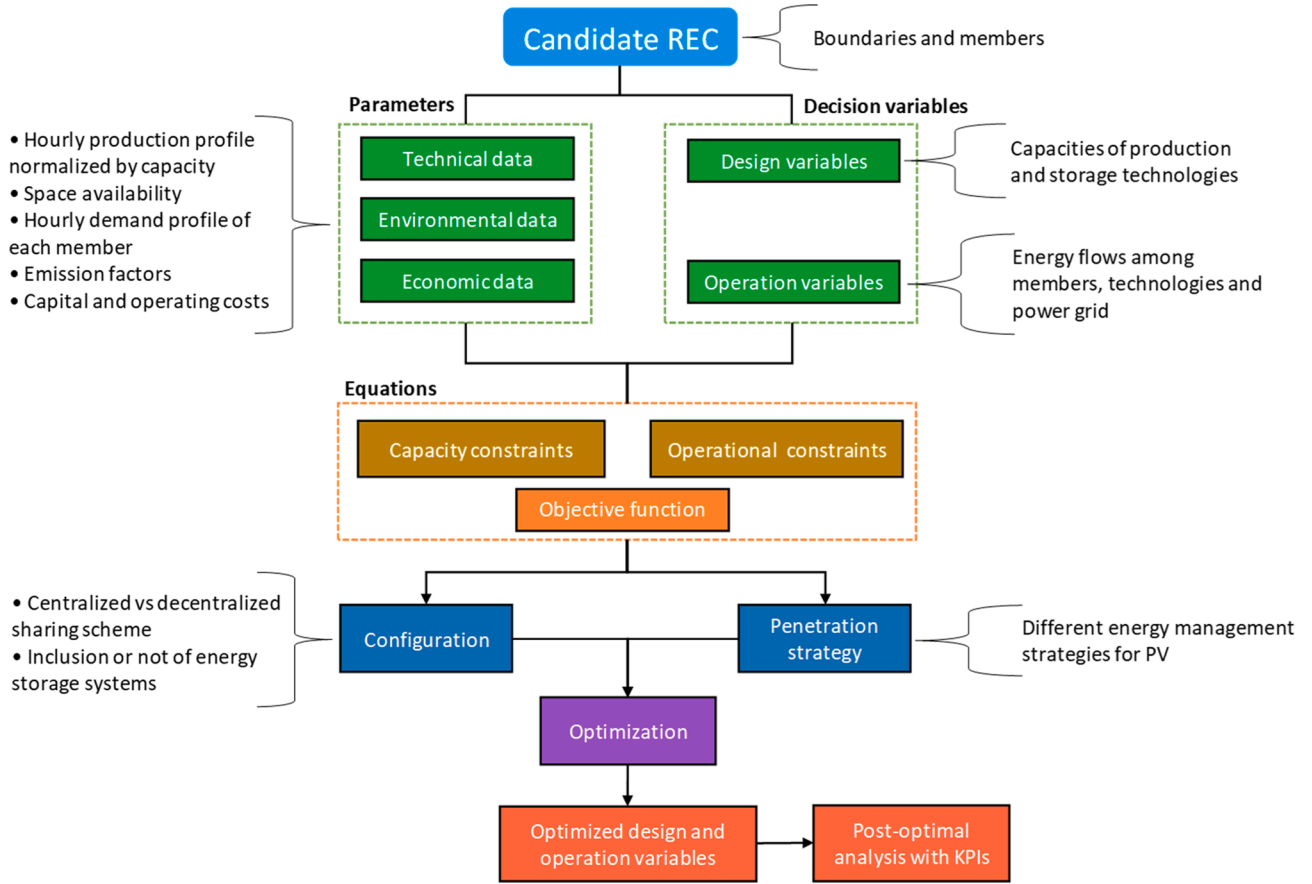


Fig. 2. Flowchart of the proposed optimization model and performance assessment.

The amount of energy used for PSC is calculated as in Eq. (6) and Eq. (7). PSC in the distributed configuration has to be calculated as the minimum, for each time-step, between the total energy production and the energy demand of each building, as formulated in Eq. (6). For the centralized case, PSC is equal to zero being all members not physically connected to any production technology, as expressed in Eq. (7).

Distributed configuration

$$EN_{tb}^{PSC} = \min\{EN_{tb}^{prod}, DEM_{tb}^{en}\} \quad \forall t, b \quad (6)$$

Centralized configuration

$$EN_{tb}^{PSC} = 0 \quad \forall t, b \quad (7)$$

The energy surplus for each member b in the distributed case, or for the whole REC in the centralized configuration, is calculated with Eq. (8) and Eq. (9), respectively. It is defined as the exceeding energy production after self-consumption of each member b at each time-step t . It can be observed that the more energy used for PSC, the less can be accounted as surplus, thus reducing the virtual sharing potential of the community. In the centralized case, consequently, the surplus of the REC is equal to the produced energy, being $\sum_b EN_{tb}^{PSC} = 0$

Distributed configuration

$$EN_{tb}^{surplus} = EN_{tb}^{prod} - EN_{tb}^{PSC} \quad \forall t, b \quad (8)$$

Centralized configuration

$$EN_t^{surplus} = EN_t^{prod} - \sum_b EN_{tb}^{PSC} \quad \forall t \quad (9)$$

On the contrary, the amount of energy still requested by each member of the REC at time step t can be defined as in Eq. (10), calculating the residual demand of member b at time t after PSC in the distributed configuration.

Distributed configuration

$$EN_{tb}^{deficit} = DEM_{tb}^{en} - EN_{tb}^{PSC} \quad \forall t, b \quad (10)$$

For the sake of brevity, the equation that applies to the centralized configuration is not provided but can be easily calculated, being $\sum_b EN_{tb}^{PSC} = 0$ (9). It should be noted that energy surplus and energy deficit are here defined as auxiliary design variables to gain a higher understanding of how released and drawn energy to and from the grid is calculated from the model. Indeed, released and drawn energy are the sole quantities effectively measured by the REC meters, from which economic incentives will be assigned to the REC. More precisely, released and drawn energy make sense only at the community level, as expressed in the balances of Eq. (11) and Eq. (12) for the distributed configuration, and Eq. (13) and Eq. (14) for the centralized one.

Distributed configuration

$$EN_t^{released} = \sum_b EN_{tb}^{surplus} \quad \forall t, b \quad (11)$$

$$EN_t^{drawn} = \sum_b EN_{tb}^{deficit} \quad \forall t, b \quad (12)$$

Centralized configuration

$$EN_t^{released} = EN_t^{surplus} \quad \forall t \quad (13)$$

$$EN_t^{drawn} = EN_t^{deficit} \quad \forall t \quad (14)$$

For both configurations, the amount of energy shared under the VSC scheme is calculated as in Eq. (15). According to the Italian regulation, the virtually shared energy is set equal to the minimum between the energy drawn from the main grid and the energy released to the grid on an hourly basis.

$$EN_t^{VSC} = \min\{EN_t^{\text{released}}, EN_t^{\text{drawn}}\} \quad \forall t \quad (15)$$

The energy that is exported to and imported from the national power grid, net of VSC, is calculated as in Eq. (16) and Eq. (17), respectively. These terms strictly refer to the energy flows that are imported and exported and therefore, have to be distinguished from the VSC amounts. The formulation of these balances is fundamental to assess the economic and environmental impacts related to the energy exchanges between the national power grid and the REC.

$$EN_t^{\text{exp}} = EN_t^{\text{released}} - EN_t^{VSC} \quad \forall t \quad (16)$$

$$EN_t^{\text{imp}} = EN_t^{\text{drawn}} - EN_t^{VSC} \quad \forall t \quad (17)$$

Energy storage systems (ESSs)

Equations, constraints and balances referring to the installation of ESSs are here presented. When considering ESSs the following assumptions have been made:

1. In the distributed case, each ESS can exchange energy only with REC members that are directly connected to production technologies, i.e., roof-PV.
2. ESSs cannot withdraw energy from the national power grid during charging cycles.
3. Charging and discharging of the ESSs cannot occur within the same time step.

Distributed configuration

For each time-step t , the constraints for maximum and minimum energy that can be stored within the ESSs are calculated as in Eq. (18) and (19), respectively. The term dod represents the depth of discharge of the battery and is expressed as a percentage of the total capacity:

$$SoC_{tb} \leq \text{cap}_b^{\text{ESS}} \quad \forall t, b \quad (18)$$

$$SoC_{tb} \geq \text{cap}_b^{\text{ESS}} * (1 - \text{dod}) \quad \forall t, b \quad (19)$$

The maximum amount of energy that can be discharged by the ESS at each time step t is calculated as in Eq. (20). Similarly, Eq. (21) poses an upper bound on the energy used for charging the ESS within each time-step t .

$$EN_{tb}^{\text{discharge}} \leq p^{\text{ESS}, \text{max}} \quad \forall t, b \quad (20)$$

$$EN_{tb}^{\text{charge}} \leq p^{\text{ESS}, \text{max}} \quad \forall t, b \quad (21)$$

Eq. (22) expresses the dynamic of the state of charge (SoC_{tb}) of each ESS considering the efficiency of charging and discharging cycles, η^{in} and η^{out} .

$$SoC_{tb} = SoC_{t-1, b} + EN_{tb}^{\text{charge}} \cdot \eta^{\text{in}} - \frac{EN_{tb}^{\text{discharge}}}{\eta^{\text{out}}} \quad \forall b, t > 1 \quad (22)$$

When ESSs are implemented within the REC, Eq. (8) and Eq. (9) should be replaced by the following two equations. Part of the energy surplus can be stored in ESSs, whilst part of the energy deficit can be covered by discharging energy from ESS.

$$EN_{tb}^{\text{surplus}} = EN_{tb}^{\text{prod}} - EN_{tb}^{\text{PSC}} - EN_{tb}^{\text{charge}} \quad \forall t, b \quad (23)$$

$$EN_{tb}^{\text{deficit}} = \text{DEM}_{tb}^{\text{en}} - EN_{tb}^{\text{PSC}} - EN_{tb}^{\text{discharge}} \quad \forall t, b \quad (24)$$

Finally, total energy accounted as PSC is now calculated as in Eq. (25). ESSs increase the PSC, permitting the consumption of the stored energy when production within the specific time step is not enough to cover the energy demand of the REC members.

$$EN_{tb}^{\text{PSC, ESS}} = EN_{tb}^{\text{PSC}} + EN_{tb}^{\text{discharge}} \quad \forall t, b \quad (25)$$

Centralized configuration

For the centralized case, Eqs. (18) - (22) can be re-used by omitting

index b , being only a centralized ESS installed within the REC. Differently from the distributed case, the energy exported to and imported from the national power grid is calculated by replacing Eq. (16) and Eq. (17) with Eq. (26) and Eq. (27), respectively.

$$EN_t^{\text{exp}} = EN_t^{\text{released}} - EN_t^{VSC} - EN_t^{\text{charge}} \quad \forall t \quad (26)$$

$$EN_t^{\text{imp}} = EN_t^{\text{drawn}} - EN_t^{VSC} - EN_t^{\text{discharge}} \quad \forall t \quad (27)$$

Finally, Eq. (28) expresses the energy shared under the VSC scheme when a centralized ESS is installed.

$$EN_t^{\text{VSC, ESS}} = EN_t^{VSC} + EN_t^{\text{discharge}} \quad \forall t \quad (28)$$

Objective function

The objective function is built to maximize the NPV of the REC over a chosen investment horizon.

$$\max\{NPV\} = -INV + \sum_y \frac{CF_y}{(1 + \text{disc})^y} \quad (29)$$

Investment costs are calculated considering capital expenditures for the installation of PV and, eventually, ESSs.

$$INV = \sum_b (PWR_b \cdot \text{capex}^{\text{prod}}) + \text{EES} \cdot \text{capex}^{\text{ESS}} \quad (30)$$

Cash flows are supposed to be equal for each operating year within the investment horizon and can be calculated as:

$$CF_y = SAV_y + REV_y - \text{COST}_y \quad (31)$$

REC members owning PV panels yield economic benefits thanks to savings in their energy bills achieved with PSC, as shown in Eq. (32). In the centralized configuration, savings are equal to zero for all members (being PSC absent). It has to be highlighted that VSC is not comprehended in the formula. Shared energy is drawn by the power grid; therefore, it is bought at the unitary cost from the national electricity market, thus not increasing energy savings. If ESSs are installed, then the term EN_{tb}^{PSC} must be replaced by $EN_{tb}^{\text{PSC, ESS}}$.

$$SAV_y = \sum_t \sum_b EN_{tb}^{\text{PSC}} \cdot \text{buy}^{\text{grid}} \quad (32)$$

Revenues of the REC are calculated as in Eq. (33). VSC is firstly sold to the power grid at the unitary price of $\text{sell}^{\text{grid}}$, before being virtually consumed by the REC members. Eq. (33) suggests that, under the Italian regulatory framework, VSC is associated with the highest economic benefit. If ESSs are installed, then the term $EN_t^{\text{VSC, ESS}}$ must replace EN_t^{VSC} .

$$REV_y = \sum_t (EN_t^{\text{VSC}} \cdot (\text{inc} + \text{ncr} + \text{sell}^{\text{grid}}) + EN_t^{\text{exp, export}} \cdot \text{sell}^{\text{grid}}) \quad (33)$$

Finally, operational costs are expressed in Eq. (34). It is worth noting that energy drawn from the power grid, EN_t^{drawn} , and bought at the unitary value buy^{grid} , is not considered. Otherwise, this cost would be accounted for twice in the objective function, being already considered in savings with Eq. (32).

$$\text{COST}_y = \sum_b PWR_b \cdot \text{opex}^{\text{prod}} + \sum_b \text{cap}_b^{\text{ESS}} \cdot \text{opex}^{\text{ESS}} \quad (34)$$

3.1. Economic, environmental, and social analysis

From the definition of REC, it comes clearly out that the primary purpose is to achieve economic, environmental, or social community benefits [1]. To assess to what extent these benefits are effectively granted to members of the REC, quantitative indicators may be formulated and performances measured. Specifically, the total self-consumption index (TSCI) and the total self-sufficiency index (TSSI) have been selected to evaluate energetic and technical aspects [38]. The indicators TSCI and TSSI derive from the literature and can be

considered as two of the most used load matching indicators [39], being they able to evaluate to what extent on-site supply and demand match in time and size [40]. In detail, TSCI describes the amount of produced energy that is self-consumed, whilst TSSI measures if PV panel production covers the REC needs, reported in Eq. (35) and Eq. (36), respectively. Due to its formulation, TSSI can be also intended as the degree of independence from the national power grid. In this study, the indicators refer to a “total” self-consumption to underline the combination of both PSC and VSC, according to the Italian normative. It is worth noting that, by using both PSC and VSC for the calculation of the indicators TSCI and TSSI, comparisons among different RECs (regardless of their configuration, i.e. centralized, distributed, or hybrid) can be performed. Environmental performances are evaluated considering the avoided carbon dioxide emission index, $CO_{2, \text{avoided}}$, calculated as the percentage of CO_2 emissions avoided thanks to the constitution of the REC. When a REC is constituted, the energy produced by renewable technologies reduces the dependence on the grid. In this paper, a net-balancing approach to carbon dioxide emissions is chosen, and, therefore, VSC is considered carbon neutral. Differently, a national-specific standard emission factor is associated with the energy imported from the grid (different from the physically or virtually self-consumed energy). From the social viewpoint, RECs are candidate communitarian solutions to reduce energy poverty affecting vulnerable demographic groups. According to the Energy Poverty Observatory (EPO), energy poverty occurs when energy bills represent a high percentage of consumers’ incomes or when consumers are forced to reduce the energy consumption of their households, affecting physical and mental health and well-being [41]. From the data shared by EPO, over 34 million European citizens are experiencing energy poverty and, in the specific case of Italy, 14.1 % of households are unable to keep their homes adequately warm [41]. The proposed new expenditure-based indicator is called energy poverty help (EPH), and is calculated as in Eq. (37). It can be read as the number of families in energy poverty conditions that can be financially helped thanks to the distribution of REC revenues. Exported energy to the grid is not linked to self-consumption, both virtually or physically, and can be sold in the free electricity national market. Therefore, economic benefits associated with the annual exported energy could be used to contribute to the annual electrical energy expenditures of a typical Italian family with three members considering as a reference the annual energy demand of 2700 kWh/y according to the Italian National Institute of Statistics [42]. Under these hypotheses, the EPH index can be calculated as in Eq. (38).

$$TSCI = \frac{PSC + VSC}{PROD} = \frac{\sum_{t,b} EN_{tb}^{PSC} + \sum_t EN_t^{VSC}}{\sum_{tb} EN_{tb}^{prod}} \quad (35)$$

$$TSSI = \frac{PSC + VSC}{DEM} = \frac{\sum_{t,b} EN_{tb}^{PSC} + \sum_t EN_t^{VSC}}{\sum_{tb} DEM_{tb}} \quad (36)$$

$$CO_{2, \text{avoided}} = \left[\frac{\sum_{t,b} EN_{tb}^{PSC} + \sum_t EN_t^{VSC}}{\sum_{tb} DEM_{tb}} + \frac{\sum_t EN_t^{prod} \cdot lcef^{pv} + \sum_t EN_t^{discharge} \cdot lcef^{ess}}{\sum_{tb} DEM_{tb}^{en} \cdot sell^{grid}} \right] \cdot 100 \quad (37)$$

$$EPH = \frac{\sum_{t=1}^{8760} EN_t^{exp} \cdot sell^{grid}}{2700 \text{ buy}^{grid}} \quad (38)$$

4. Case study

The optimization model presented in Section 3 is applied to an eligible REC in Catania, Sicily, depicted in Fig. 3. The neighborhood comprehends 17 buildings, whose final use is mostly residential.

The main characteristics of the buildings of the area have been listed in Table 1. Building members are identified in terms of available roof surfaces, number of inhabitants, and dwelling units; these data have been reported in columns 2 to 4, respectively. Hourly electrical consumption profiles for a yearly time horizon have been derived from the U.S. Department of Energy (DOE) database, which includes hourly and sub-hourly load data for several types of buildings located in different climatic regions [43]. For this study, the region whose climate was more similar to that of the case study has been selected, as suggested in Viti et al. [6]. Afterwards, hourly data for a single-family building have been scaled using the average yearly consumption of a typical Italian family of three members as a reference, i.e. 2,700 kWh/year [42].

Regarding renewable production systems, PV panels have been selected for the affordable investment costs, practicable installation on building roofs, and high-capacity factors in Mediterranean areas. Technical constraints on the maximum allowable installed size for PV panels are given by considering the total available roof of each building member. PV can be installed on all buildings, but considering variable surfaces. Hourly PV production profiles have been computed in PVGIS [44]. For this study, crystalline silicon cell panels have been chosen,



Fig. 3. Neighborhood selected as a case study, Catania.

Table 1

Main characteristics of each member of the REC.

Building id	Available tilted roof facing south [m ²]	No. of inhabitants	No. of dwelling units
1	31.00	64	21
2	32.00	60	19
3	28.00	28	9
4	31.00	30	10
5	30.00	28	8
6	28.00	27	9
7	22.00	23	9
8	23.00	29	9
9	20.00	24	5
10	21.00	26	6
11	22.00	25	8
12	20.00	9	3
13	11.00	6	2
14	9.00	5	2
15	240.00	4	1
16	238.00	3	1
17	332.00	3	1

accounting for system losses equal to 14 % [44]. Each collector is installed with azimuth and tilt respectively equal to 30° and 2°, to maximize the production yield, and with the constraint of 6 m² of area per installed kW [45]. Fig. 4 shows the monthly average daily production of a 1 kW PV system, with an annual energy yield equal to 1,776 kWh per kW of installed PV nominal power. It results that a maximum value of 187 kW_n can be installed on building roofs and this value has been set as the upper bound in the optimization model.

In this study, the two energy-sharing configurations “distributed” and “centralized” are compared for the scenarios with and without ESSs, also accounting for different strategies in terms of variable PV penetration within the REC. An overview of the sharing configurations, scenarios and strategies is presented in Fig. 5. As said, strategies #1 and #2 provide for different levels of PV penetration in the range [25 %, 50 %, 75 %, 100 %], according to which the installation of PV is prioritized for buildings members with higher energy consumption (strategy #1) or with lower energy consumption (strategy #2).

Table 2 lists the economic and technical parameters used to feed the optimization model. Economic data, such as the network charge restoration and the incentive tariff, derive from the Italian regulation on REC [2,3]. The chosen investment horizon for the computation of annual

cash flows has been set equal to 20 years, again a threshold indicated by the GSE [2]. Other cost items and technical aspects have been derived from previous studies available from the literature. As introduced in Section 2, economic benefits and costs are equally distributed among the members of the REC.

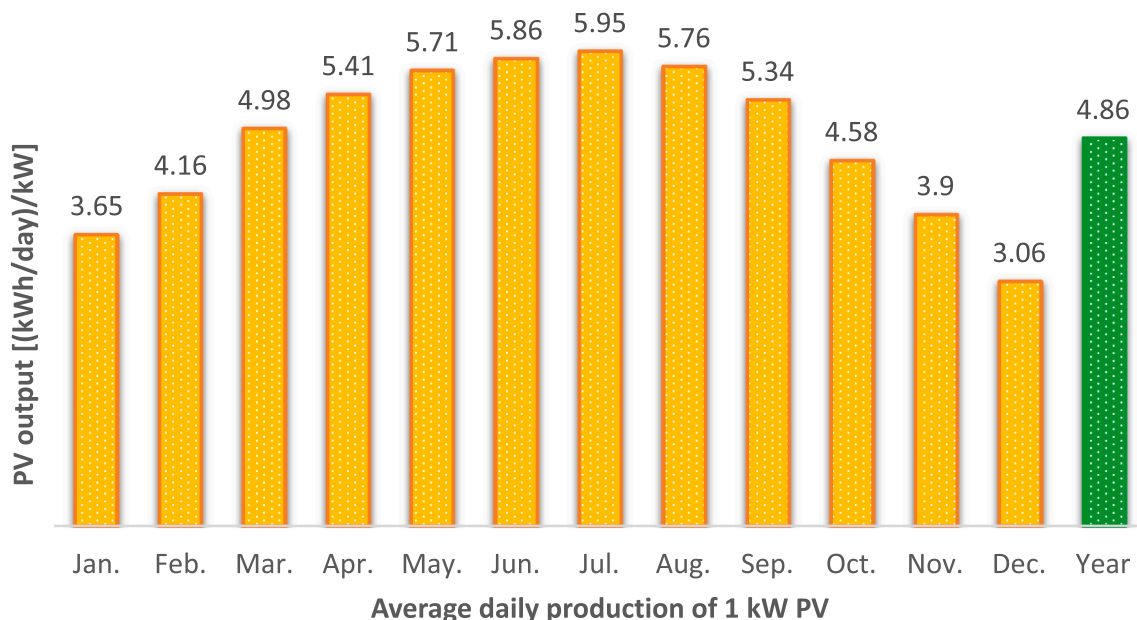
5. Results and discussion

The optimization model maximizes the NPV along the entire investment horizon of the REC. In the first place, the results for the scenarios without ESSs are presented and discussed. It is worth pointing out that the following graphs report the PV nominal power on the x-axes within the range 0 – 187 kW i.e., from the minimum to the maximum installable nominal power resulting from the available rooftop areas of buildings.

5.1. Centralized and distributed sharing configuration without ESSs

The optimized results in terms of NPV at varying PV nominal power are plotted in Fig. 6, for both the centralized and distributed energy-sharing configuration without ESSs and adopting strategy #1, i.e., prioritizing the installation of PV in buildings with higher energy demand. Then, as previously introduced, the distributed configuration is split into four different percentages of PV penetration. Red dots identify the optimal condition, namely the PV nominal power to be installed to obtain the maximum NPV over the selected investment horizon of 20 years.

As can be observed, the curves of Fig. 6 display a similar behavior: initially, at lower PV sizes, the NPV increases almost linearly. The initial increasing trend can be explained by considering that all the produced energy (in both centralized and distributed sharing configuration) is mainly used as PSC or, at least, VSC from the REC members. In this case, the economic benefits for the REC members increase, in terms of both energy savings and revenues. This trend continues for all configurations until it reaches a point of maximum for the NPV, where almost all the demand for buildings is covered by VSC and PSC. Consequently, revenues from selling exported energy to the grid increase but, in the end, cannot positively compensate the capital and operating costs for the installation of higher capacity PVs, causing the curves to have negative slopes. These considerations are also confirmed by Fig. 7, which plots the energy flows associated with economic benefits, in terms of exported

**Fig. 4.** Average daily production of a 1 kW PV, per month.

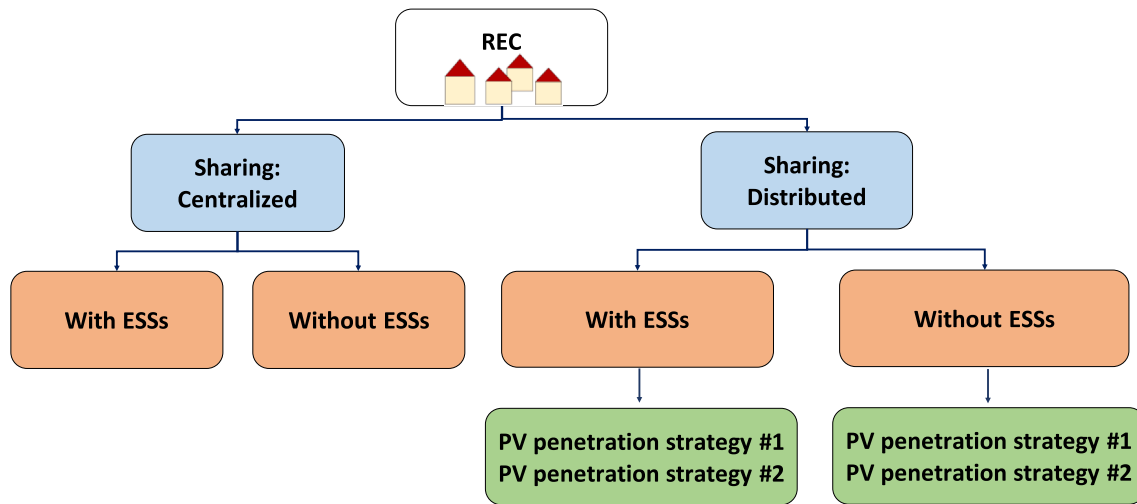


Fig. 5. Optimized REC sharing configurations, scenarios and strategies.

Table 2

Economic and technical assumptions.

Description	UoM	Value
PV CAPEX	€/kW _n	1,500 [46]
PV OPEX	(€/kW _n)/y	30 [46]
ESS CAPEX	(€/kWh)/y	1,100 [46]
ESS OPEX	€/kWh	20 [46]
Investment horizon	y	20 [2]
Discount rate	–	0.03 [4]
Unitary cost of buying electricity from the power grid	€/kWh	0.145 [46]
Unitary cost of selling electricity to the power grid	€/kWh	0.60 [46]
Network charge restoration	€/kWh	0.087 [2]
Incentives tariff	€/kWh	0.110 [2]
Charging efficiency of ESSs	–	0.95 [20]
Discharging efficiency of ESSs	–	0.95 [20]
Depth of discharge of ESSs	–	0.7 [20]

energy and total self-consumption, i.e., a combination of VSC and PSC. It is worth observing that the combination of these two contributions does not vary at varying the sharing configuration, therefore no specification in this sense is reported in Fig. 7. For each sharing configuration, the energy exported to the national power grid shows a positive convexity with respect to the size of PV panels, whilst total self-consumption (PSC + VSC) displays a negative one. These trends may be explained by considering that exported energy increases more rapidly than total self-consumption for high values of PV installed nominal power. Then, comparing the two sharing configurations in Fig. 6, it arises that the centralized configuration yields significantly higher values of NPV for any fixed value of the PV nominal power.

To explain these differences, the valorization mechanism for energy released to the grid or physically/virtually shared has to be explained. Indeed, under the Italian regulatory framework, VSC is more incentivized than PSC. As already mentioned, the total self-consumption (i.e., PSC + VSC) under the same PV nominal power is the same for each configuration, what changes is the way it is divided into PSC and VSC. For instance, PSC is not permitted in the centralized configuration and, thus, the energy produced by PV is released into the national power grid and shared by REC members according to a VSC scheme. Differently, in the distributed configurations energy from PV panels is firstly physically self-consumed (PSC), and then virtually shared among the members (VSC). Therefore, as a general principle, the amount of VSC in any distributed configuration is always smaller than in the centralized one, for each fixed installed PV nominal power, explaining the gap between the two configurations. Thus, the centralized configuration is always the more profitable being VSC more incentivized than PSC. Similarly, when

comparing the curves for 25 % and 50 % PV penetration, the decrease can be attributed to the fact that more buildings are physically connected to PV panels causing PSC to increase, and consequently, VSC to decrease.

Moving from the 25 % of PV penetration forward, the relative distance among the curves tends to decrease. Nonetheless, a different behavior can be observed when shifting from 75 % to 100 %. To explain this behavior, it can be useful to take advantage of the bar chart of Fig. 8. To allow comparison, a PV power of 107 kW_n has been chosen, which can be considered as the optimal value for both the 75 % and 100 % cases.

To evaluate the different contributions' impact on the results, the energy produced from PV has been distinguished into three terms: PSC, VSC, and exported. The share of exported energy remains the same, here equal to 33 %. Therefore, as expected, the differentiation in terms of NPV has to be charged to the variation in PSC and VSC. Indeed, as can be observed, shifting from 75 % to 100 %, the VSC increases and PSC decreases, correspondently.

Being the VSC scheme more impacting for the REC, it is reasonable to evaluate the amount of energy virtually shared among the REC members over the amount of produced energy. Fig. 9 shows the ratio between VSC and the total PV production, for the centralized and distributed configurations, considering the scenario without ESSs and strategy #1 for PV penetration. As expected, the centralized curve tops all the others since no PSC occurs, and the totality of the energy self-consumed has to be computed to VSC. In the centralized case, for small PV sizes, the total energy production accounts for VSC with a ratio equal to 1, afterwards the ratio decreases, being more energy exported to the national power grid (overproduction). Differently, the distributed configurations show an opposite trend: for smaller PV sizes energy is almost entirely used under the PSC scheme, with a disadvantage for VSC.

It is worth noting that these commented results are strongly affected by how supply and load profiles match together at both the building and community levels. This depends mainly on how PVs are physically installed among members, i.e., which strategy has been chosen during the REC constitution. Fig. 10 and Fig. 11 report the NPV values and the ratio between VSC and production for centralized and distributed configurations, under the scenario without ESSs, but choosing strategy #2, i.e., with PVs priorly installed in buildings with lower energy demands. Here, the differences among the curves are smooth. Indeed, at a fixed PV size and increasing the PV penetration, the VSC decreases and the PSC increases, due to the fact that buildings with higher energy demands are now more intensively covered with PV panels and their consumption scheme is entirely classified as PSC. The difference between centralized and distributed configuration at 25 % PV penetration is, therefore, less

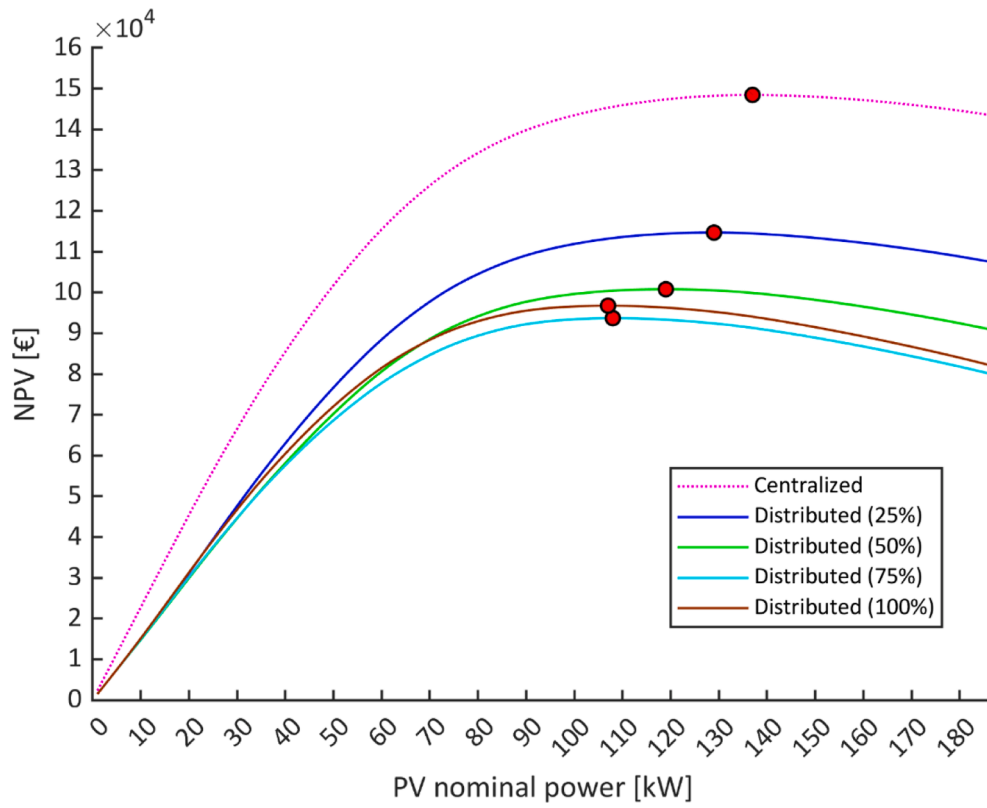


Fig. 6. NPV for the centralized and distributed sharing configurations without ESSs; strategy #1.

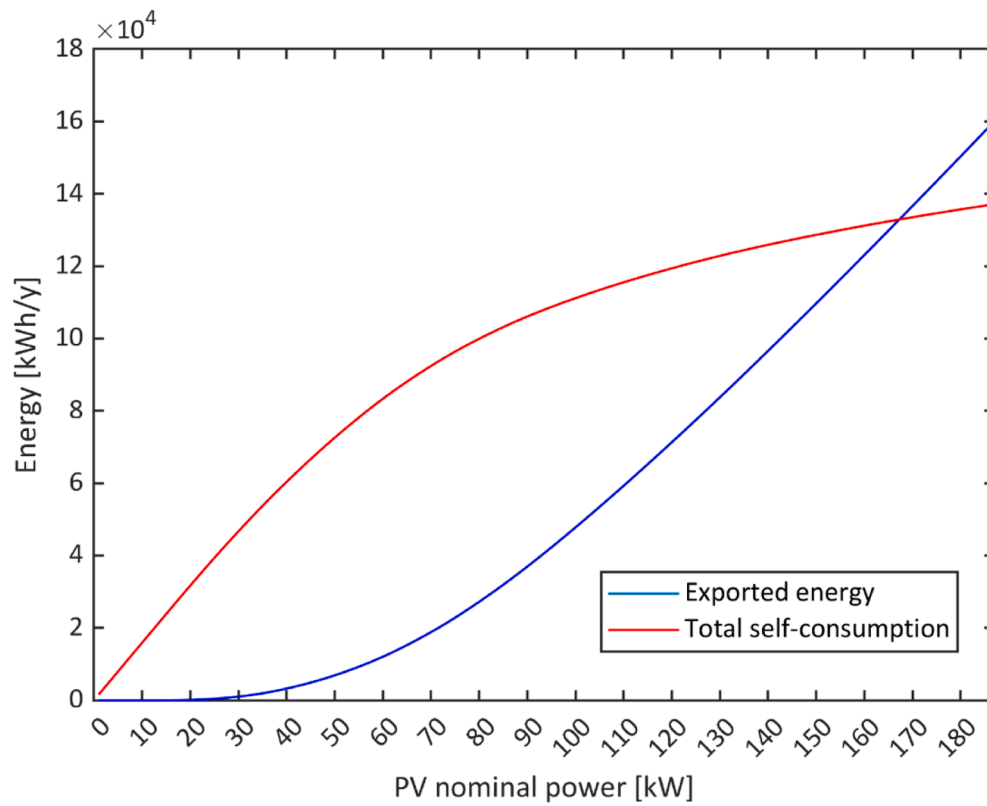


Fig. 7. Exported and total self-consumed energy without ESSs; strategy #1.

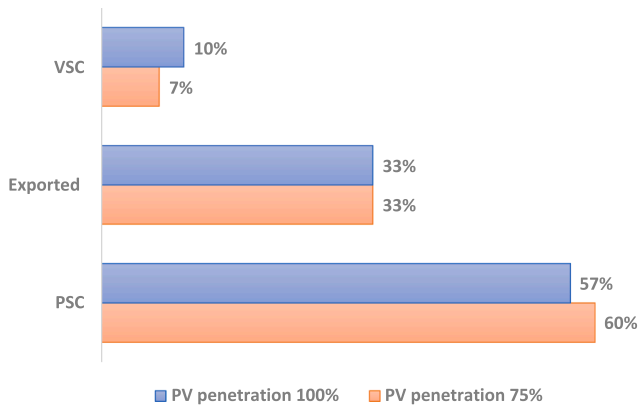


Fig. 8. PV penetration at 75 % and 100 %, scenarios without ESSs; strategy #1.

marked, given that the comparison is now made on similar values of VSC for both configurations. According to strategy #2, the curve for the 25 % PV penetration indicates that the buildings equipped with PV panels are those with lower electrical demand and, therefore, the VSC has become comparable with the centralized configuration.

The ratio between the VSC over the PV production is highlighted in Fig. 11. Here, the curves corresponding to smaller PV sizes are steeper than the obtained for strategy #1, since PSC is here less impacting with respect to the total self-consumed energy. No differences occur between strategies #1 and #2 for the centralized and 100 % distributed configurations, allowing the former to be interpreted as an upper limit in terms of NPV and VSC.

As a final remark, and although some variations, both Fig. 6 and Fig. 10 present an optimum PV size in the range of around 100 – 140 kW_n for the NPV maximization and that can be taken as a reference during the design phase of the REC, regardless of the adopted configuration.

5.2. Centralized and distributed sharing configuration with ESSs

In this second part of the section, results refer to the same configurations and scenarios set before, but including the installation of ESSs. The optimized results in terms of NPV at varying PV nominal power are plotted in Fig. 12, for both the centralized and distributed energy sharing configuration with ESSs and adopting strategy #1, i.e., prioritizing the installation of PVs in buildings with higher energy demand.

The optimized results in terms of NPV and PV nominal power vary significantly with respect to the scenarios without ESSs. As Fig. 12 shows, curves maintain a similar behavior to those without ESSs, plotted in Fig. 6, but resulting in significantly lower economic benefit for the REC members (NPV is at least 4 times smaller). Furthermore, only the centralized sharing configuration leads to positive values of the NPV, meaning that the other configurations are neither economically attractive nor competitive. A great impact on these results can be computed to the capital costs needed for the ESSs installation, as reported in Table 2. Optimal PV nominal powers for each scenario are slightly higher: in this case, indeed, overproduction is not entirely exported to the power grid but is used to charge batteries and disaggregate production and load, hence contributing also to the increase of total self-consumption, in terms of both PSC and VSC. Thus, how production and ESSs are physically distributed among members strongly affects the performance of the REC.

Fig. 13 reports the ratio between VSC and production at varying the PV nominal power for the centralized and distributed configurations when ESSs are included, and considering strategy #2. The ESS has been chosen as fully charged at the time $t = 1$, explaining the ratio higher than the unity when small PV sizes are considered in the centralized case. It is worth noting that the centralized curve decreases with a smaller slope with respect to the scenario without ESSs in Figure 9, suggesting that ESSs allow for a bigger amount of energy to be accounted as VSC among REC members. On the contrary, when ESSs are distributed within the REC, energy surpluses are primarily used to

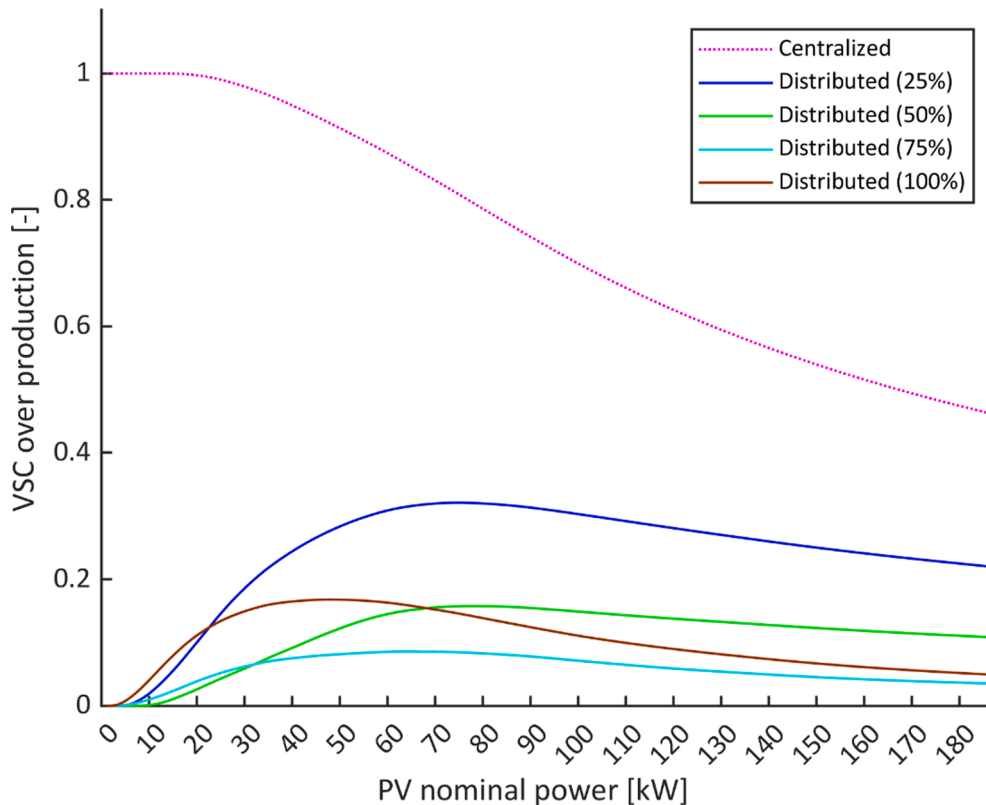


Fig. 9. VSC over total production for the centralized and distributed configurations without ESSs; strategy #1.

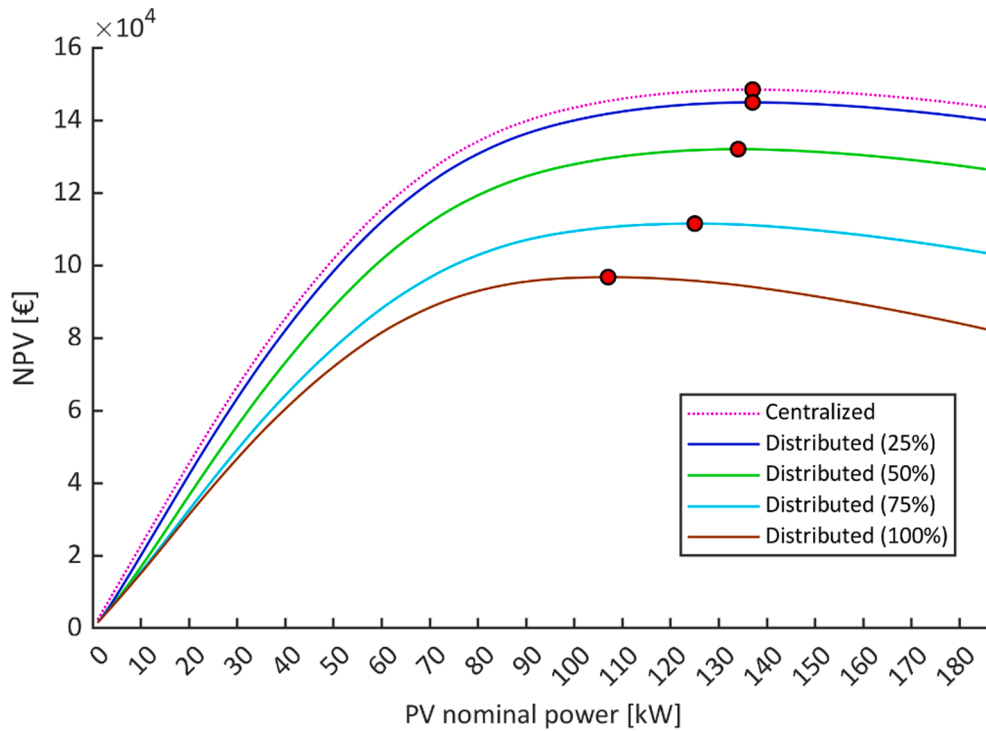


Fig. 10. NPV for the centralized and distributed sharing configurations without ESSs; strategy #2.

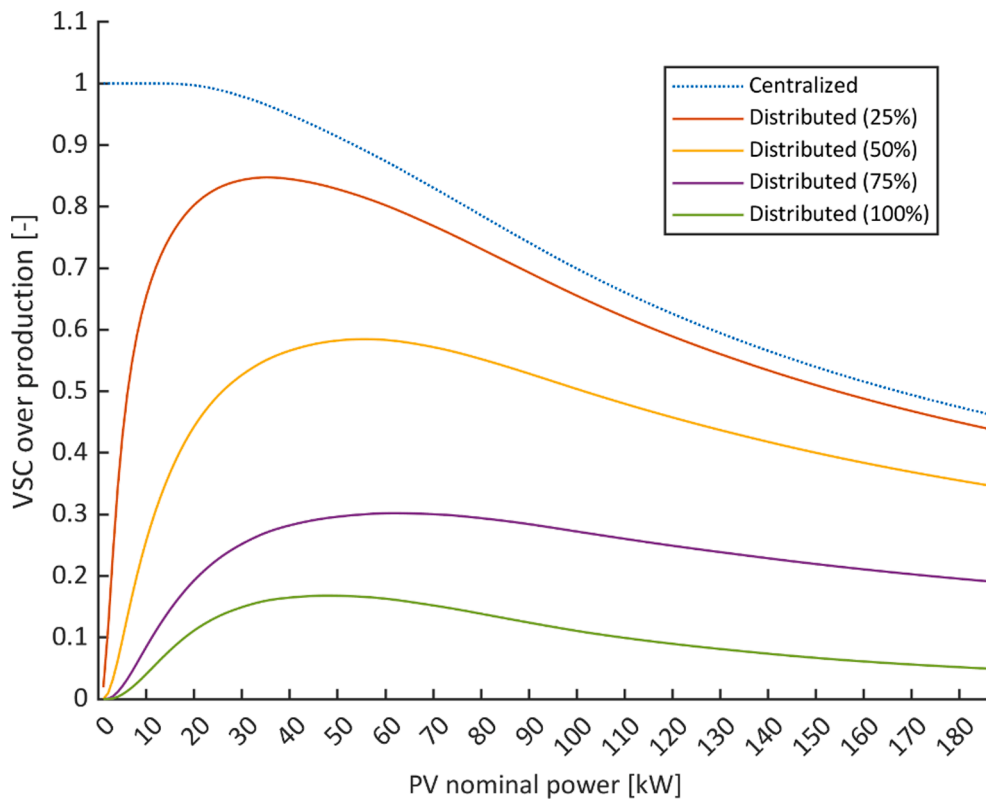


Fig. 11. VSC over total production for the centralized and distributed configurations without ESSs; strategy #2.

charge batteries and, only afterward, the exceeding production is virtually shared among REC members. As a consequence, the curves for distributed sharing configurations are flatter if compared to the scenario without ESSs, suggesting that less energy is virtually shared among REC members, specifically less than 10 % of total produced energy for three

of the four distributed configurations.

Similar results can be achieved for the centralized and distributed configurations with ESSs accounting for strategy #2.

The bar charts in Fig. 14 graphically summarize the difference between the scenarios without and with ESSs and choosing the centralized

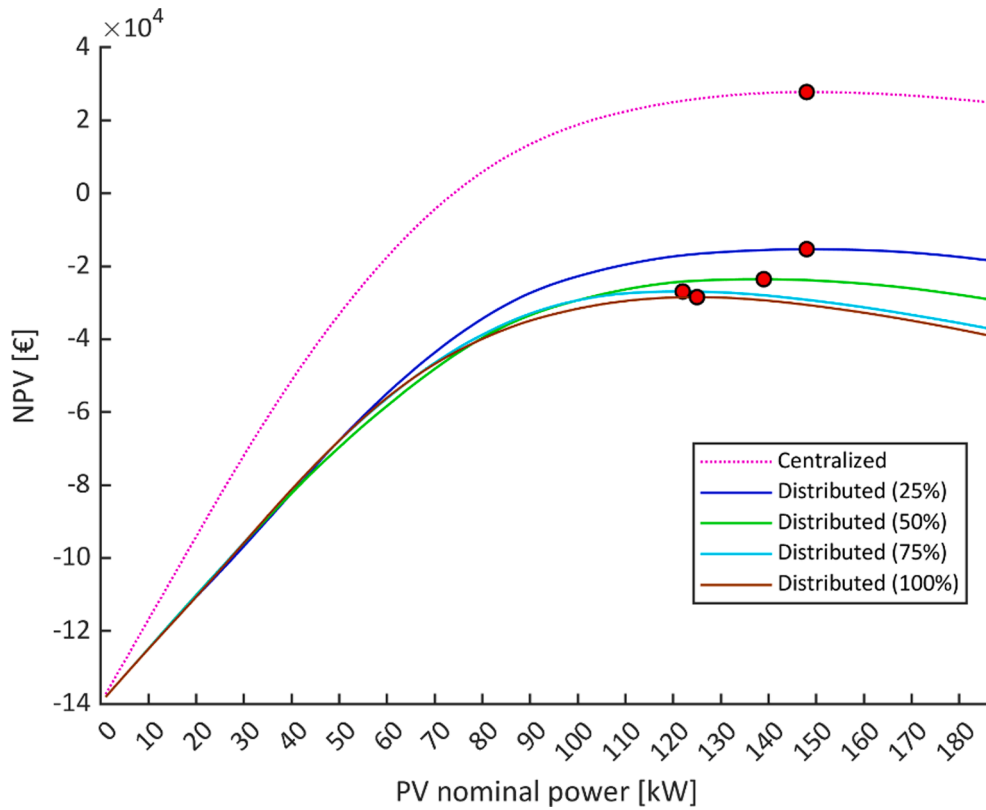


Fig. 12. NPV for the centralized and distributed sharing configurations with ESSs; strategy #1.

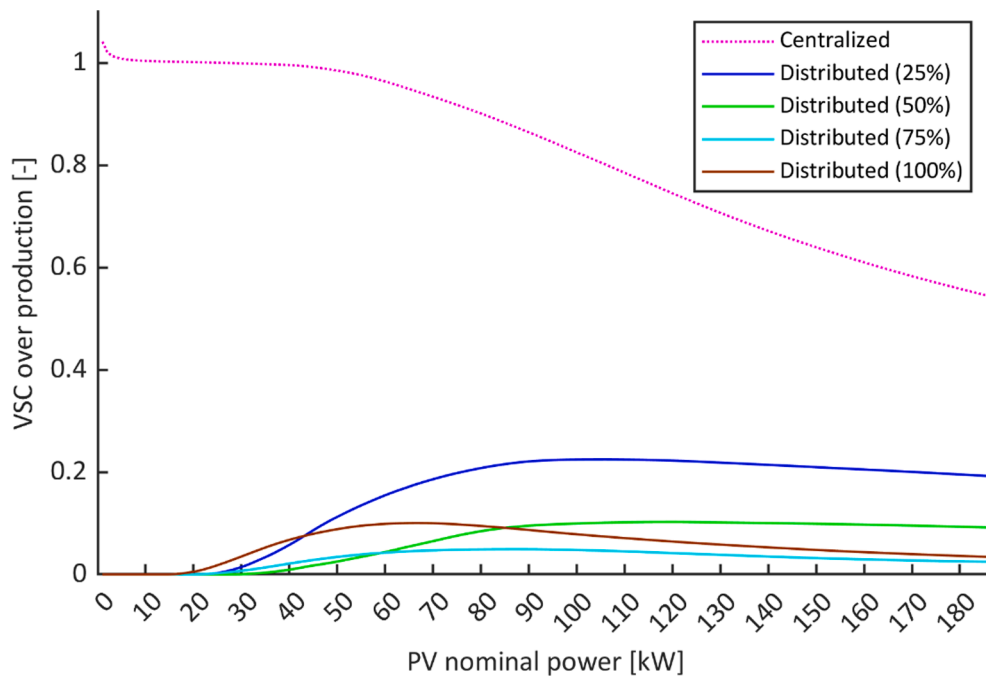


Fig. 13. VSC over total production for centralized and distributed configurations with ESSs; strategy #1.

and distributed sharing configurations. The case for 25 % PV penetration has been selected for discussion, due to the higher NPV values and the higher ratio between VSC and production. To allow comparison, the optimal PV power of 137 kW_n has been selected.

From the comparison between the two centralized configurations, without and with ESSs, plotted in Fig. 14(a), it emerges that the amount

of energy exported to the grid diminishes, favoring the amount of VSC among the REC members. On the other side, the distributed configuration, reported in Fig. 14(b), allows an increase in terms of PSC, thus impacting on the virtually shared energy, i.e., VSC. This refers also to the economic benefits of the REC, being the VSC the incentivized scheme by the Italian regulation. In both cases, however, the total self-consumption

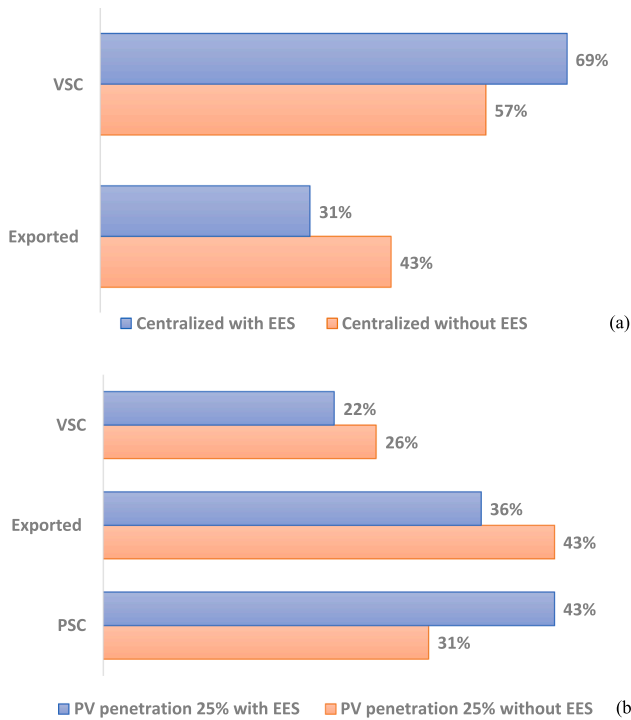


Fig. 14. Comparison between centralized configuration without ESSs and with ESSs (a) and 25 % distributed configuration without ESSs and with ESSs (b).

of the REC, namely VSC + PSC, increases. It is worth noting that comparable results can be observed under strategy #2, here not reported for the sake of brevity.

Economic, environmental and social performances of the REC

The key performance indicators TSCI and TSSI (for the economic evaluation), CO₂ (for carbon emissions reduction), and EPH (for social impact) introduced in Section 3 have been calculated for the chosen scenarios with and without ESSs and strategy #2, considering the optimum value for installed PV nominal power of 137 kW_n to allow meaningful comparisons. The results of the calculations are listed in Table 3. For a more accurate computation, the life cycle emission for PV panels and ESSs have been associated with the emission factors $l_{\text{cef}}^{\text{PV}} = 50 \text{ gCO}_2/\text{kWh}$ [47] and $l_{\text{cef}}^{\text{ESS}} = 175.0 \text{ gCO}_2/\text{kWh}$ [48], respectively. A standard emission factor for the energy drawn from the power grid has been chosen to be equal to $276.3 \text{ gCO}_2/\text{kWh}$ [49].

Considering the energy-related KPIs, i.e. TSSI and TSCI, the performances in terms of energy flow utilization can be discussed. As expected, the introduction of ESSs allows the disaggregation between load and supply, thus increasing the amount of energy that can be self-consumed. Similarly, they allow to achieve a higher level of independence from the power grid, assessed by the TSSI, and therefore a consequent reduction in the CO₂ emissions. Differently, the social impact decreases when centralized ESSs are installed. In fact, revenues from exported energy are balanced by the investments for installing ESSs, explaining why a minor number of families can be financially helped. In this sense, investors and, generally, stakeholders, are in an ambivalent position, being forced to choose between pursuing the maximization of energy-related performance or paving the way for acts of solidarity towards citizens living in

Table 3

KPIs for the centralized configuration with and without ESS.

KPI	Centralized without ESS	Centralized with ESS
TSSI [-]	32 %	38 %
TSCI [-]	57 %	68 %
CO ₂ , avoided [-]	22 %	34 %
EPH [#families/y]	14	10

situations of distress. Thus, it can be observed how these results become extremely important during the REC design phase, given the impact on final performances. Moreover, assessing performances with KPIs able to capture different dimensions help decision-makers to gain awareness of the sustainability assessment of the REC and to select a trade-off solution. Similar considerations apply to the distributed case.

6. Conclusions

In this study, the optimal design and operation of a REC under the Italian Regulatory Framework is investigated, taking into consideration variable distribution schemes and variations in the installing capacities of renewable production technologies. In particular, centralized and distributed schemes for PVs and storage technologies are optimized with the aim of maximizing the NPV over a specified investment horizon. In accordance with the Italian regulation, two energy-sharing configurations, namely distributed and centralized, are investigated and four purpose-built KPIs have been used to assess the impact of the optimized REC from the economic, environmental and social viewpoints. The model and the performance assessment have been studied for a candidate REC in Sicily. The main outcomes of this work can be summarized as follows:

- The centralized scenario is always the more profitable, being virtually shared energy more incentivized than physical self-consumption. As a point of discussion, it could be stated that, if it would be necessary to install production technologies physically connected to some of the buildings of the REC, while aiming at maximizing economic benefits, then it would be advisable to select the least energy-intensive buildings rather than the most energy-intensive ones.
- The installation of ESSs would improve the rates of total self-consumption of the REC, but it would also negatively affect cash flows for the high impact deriving from higher investment and maintenance costs. Comparing the KPIs for the two configurations – centralized without and with ESS – reveals that ESSs provide a higher degree of power grid independence, as measured by the higher value of the TSSI. However, because of greater investment in technologies, the beneficial social impact on REC's participants is reduced. This last remark demonstrates that environmental and social performances are at odds, and further analysis is required to determine how these two factors interact.

Overall, this work reinforces the effectiveness of the optimization approach for tackling the design and operation phases of RECs. In addition, the model and the related KPIs could support different stakeholders involved in the decision-making processes prior to the constitution of a REC and after, to monitor performances. Further directions of this study could be concentrated on the definition of a multi-objective model to solve conflicting results in terms of economic and environmental/social performances of the REC, as well as the application of the model to other national frameworks to account for local incentives or regulations and measure their impact in terms of sustainability assessment.

CRedit authorship contribution statement

Emanuele Cutore: Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing, Investigation. **Rosaria Volpe:** Data curation, Conceptualization, Methodology, Investigation, Writing – original draft, Writing – review & editing. **Roberta Sgroi:** Software, Investigation. **Alberto Fichera:** Conceptualization, Methodology, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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